Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/17/2025
PI Name:	Wu, Honglu Ph.D.		
Project Title:	DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
Start Date:	01/01/2016	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Final report date changed to 09/30/2025 per NASA JSC. (Ed., 2/18/25) NOTE: End date changed to 9/30/2023 per NASA JSC. (Ed., 11/18/21) NOTE: End date changed to 9/30/2022 per PI (Ed., 10/6/20) NOTE: Extended to 3/31/2021 per PI (March 2019)--Ed., 10/3/19		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feiveson, Alan Ph.D. (NASA Johnson Space Center) Zhang, Ye Ph.D. (NASA Kennedy Space Center) Moreno-Villanueva, Maria Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: This is an integrated project consisting of Dr. Brian Crucian's "Functional Immune Alterations, Latent Herpesvirus Reactivation, Physiological Stress, and Clinical Incidence Onboard the International Space Station" directed research; and Dr. Richard Simpson's "The Impact of an ISS Mission on the Anti-Viral and Functional Properties of NK-cells, T-cells, B-cells and Dendritic Cells," Dr. Kanokporn Rithidech's "Effects of Space Flights on the Proteome of Astronauts' Plasma," and Dr. Honglu Wu's "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" solicited research. The unique radiation environment encountered in space has been known to cause deleterious effects in humans, and these effects are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. DNA lesions, namely single- and double-strand breaks, are the most common of these effects and the most likely to compromise health. In addition to radiation, astronauts in space experience microgravity and other stress factors. In human cells, immune cells in particular, stress-induced catecholamine and corticoid release, as well as increased levels of inflammation, are known to induce DNA damage and impair DNA repair, which in turn results in single and double-strand breaks. Intact DNA repair machinery is crucial for immune cell functionality. Therefore understanding the interaction between DNA damage response and immune response is an important step towards improving knowledge of the mechanisms associated with immune dysfunction in the space environment. In this study, we propose to quantify DNA damage in astronauts' lymphocytes by measuring the frequency of single- and double-strand breaks. We will also perform RNA-seq analysis with a focus on pathways involved in DNA damage response and/or inflammatory response. Measures of DNA damage obtained before, during, and after long-duration International Space Station (ISS) missions will then be compared with the mRNA levels. The outcomes in the present study will also be compared to the measured T cell function, cytokine profiles and viral reactivation in the Integrated Immune Flight Study, which is the parental project of our proposed study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The unique radiation environment encountered in space may cause deleterious effects in humans, which are of particular concern for prolonged missions beyond the protective terrestrial magnetosphere. Space radiation-induced DNA lesions, including single- and double-strand breaks, are among the most common forms of damage and are most likely to lead to adverse health consequences. In addition to radiation exposure, astronauts experience microgravity and other stressors during spaceflight. In human cells – particularly immune cells – stress-induced catecholamines and corticosteroids, as well as increased levels of inflammation, are known to induce DNA damage and impair DNA repair, resulting in DNA single- and double-strand breaks. An intact DNA repair machinery is crucial for proper immune cell function. Therefore, understanding the relationship between the DNA damage response and immune function is essential for elucidating the mechanisms underlying immune dysfunction in the space environment. In this study, blood samples from eleven crewmembers were collected before, during, and after International Space Station (ISS) missions. Transcriptomic and microRNA analyses were performed in isolated peripheral blood mononuclear cells (PBMCs) using RNA sequencing. Differentially expressed genes (DEGs) and microRNAs during spaceflight were identified by comparing inflight samples with preflight samples. In addition, DNA damage in astronauts' lymphocytes was quantified by measuring the frequency of single- and double-strand breaks. Through this study, we made the following key observations: 1. We propose that mitochondrial dysfunction in T cells during spaceflight is caused by persistent modulation of mechanosensing receptors under microgravity conditions, triggering signaling cascades that lead to mitochondrial calcium overload. The response of PBMCs in space exhibits features of T-cell exhaustion, which are more likely initiated by microgravity rather than infection. 2. We suggest that T cells from younger astronauts respond more strongly to the space environment than those from older astronauts, potentially resulting in greater negative immune consequences. Although sample analysis has been completed, analysis of the collected data is still ongoing. Ongoing work includes the following: 1. Completion of the microRNA analysis and submission of a manuscript reporting these results. 2. Completion of the PCR analysis and submission of a manuscript reporting these results. 3. Completion of the gene expression-chromosomal location analysis and submission of a manuscript reporting these results. 4. Completion of the DNA damage analysis using PBMC samples. 5. Comparative analysis of gene expression data from this study with other immune function measurements by a NASA Johnson Space Center (JSC) biostatistician.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Moreno-Villanueva M, Jimenez-Chavez LE, Krieger S, Ding L-H, Zhang Y, Babiak-Vazquez A, Berres M, Splinter S, Pauken KE, Schaefer BC, Crucian BE, Wu H. "Transcriptomics analysis reveals potential mechanisms underlying mitochondrial dysfunction and T cell exhaustion in astronauts' blood cells in space." Front Immunol. 2025 Jan 19;15:1512578. https://doi.org/10.3389/fimmu.2024.1512578 ; PMID: 39902046; PMCID: PMC11788081 , Jan-2025

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